University of Washington site update



DELPH-IN Summit
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Singapore

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Grammar Engineering: 567

- Ling 567 languages:
  - Bardi [bcj]
  - Chuckchi [ckt]
  - Columbian-Wenatchi [col]
  - Khmer [khm]
  - Lithuanian [lit]
  - K’iche’ [quc]
  - Chadian Arabic [shu]
  - Upper Necaxa Totonac [tku]
  - Yakima Sahaptin [yak]

2015 567 languages - mapped



lat/long data mostly from wals.info; map by batchgeo.com

2015 567 languages - cupcaked



Cupcakes by Laurie Dermer

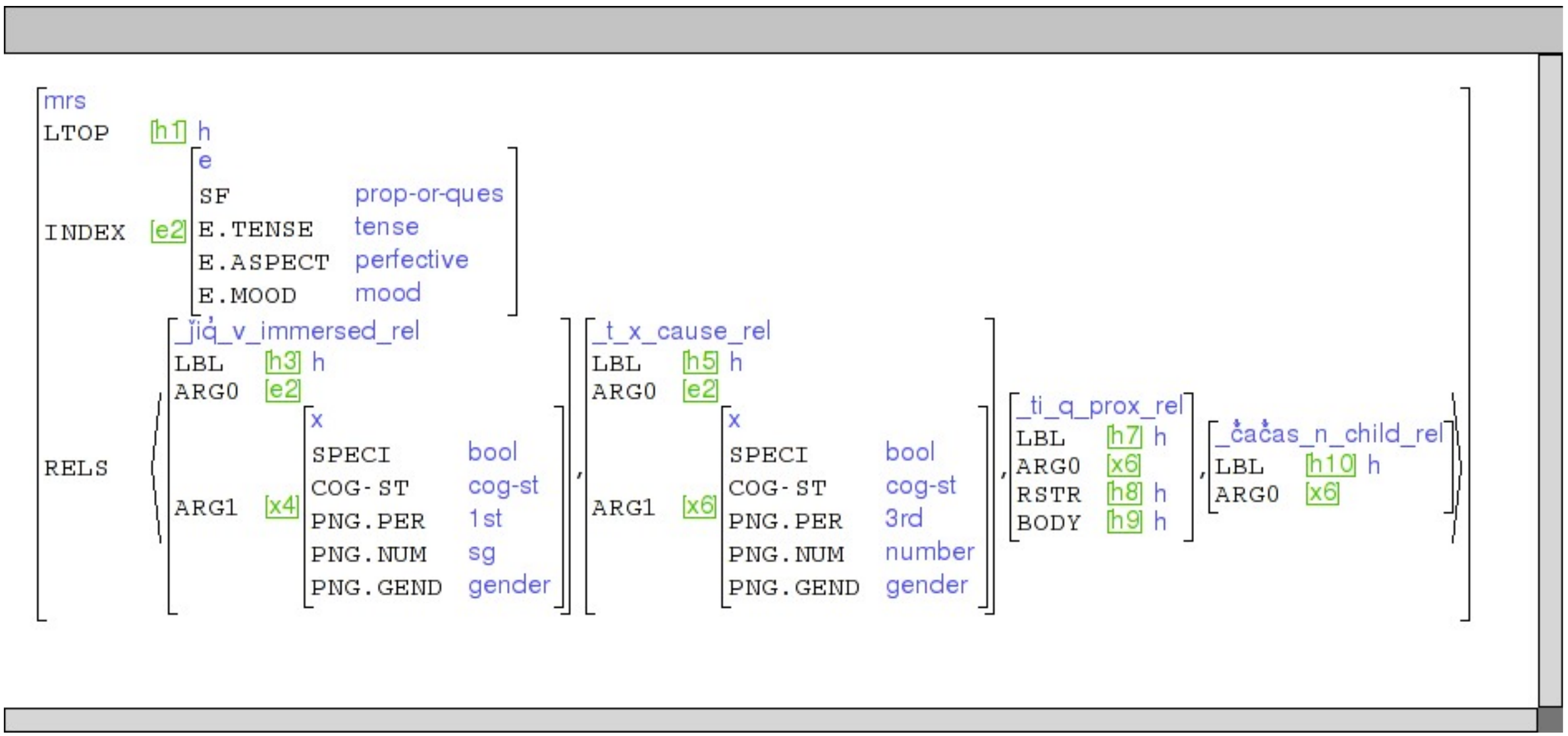
Items with end-to-end output (post transfer rule propagation)

src \ tgt	bcj	ckt	col	eng	frr	khm	lit	quc	shu	tku	yak	zak
bcj	17	17	17	17	14	15	16	14	17	17	16	14
ckt	12	13	12	13	12	12	12	9	13	11	11	11
col	16	19	21	21	14	17	20	14	20	19	19	13
eng	17	21	22	23	16	20	22	18	23	21	21	15
frr	16	17	18	18	17	16	17	12	18	16	16	14
khm	16	20	22	22	16	19	22	14	22	20	21	13
lit	18	19	20	20	16	17	19	15	19	18	18	16
quc	14	11	15	15	13	14	15	15	15	14	14	11
shu	15	18	19	21	14	18	20	15	21	20	20	14
tku	15	14	14	15	14	14	15	12	15	18	15	13
yak	16	18	19	20	14	19	19	13	16	17	18	15
zak	14	14	16	16	12	15	15	13	14	15	14	16

Grammar Engineering: Lushootseed [lut] (Joshua Crowgey)

- Dissertation prospectus: Development of Lushootseed Grammar w/CLIMB to explore syntax/semantics of multicategoricity of Lushootseed
- Initial grammar now parses small sentence templates exhibiting some interacting phenomena:
    - derived transitive verbs from intransitives (applicatives and causatives)
    - subject clitics in S2 position (treated as auxiliaries)
    - object marking suffixes
    - dropped arguments
- Lexical database exports to FST & tdl

ʔuǰiɬ ti čačas	ʔuǰiɬəd ti čačas
ʔu-ǰiɬ ti čačas	ʔu-ǰiɬ-t ti čačas
PERF-immersed DET child	PERF-immersed-ICS DET child
“The boy drowned.” [lut]	“Someone drowned the boy.” [lut]



Matsigenka [mcb] and Nuu-chah-nulth [nuk] (David Inman), Thai (Glenn Slayden)

- Matsigenka: Corpus analysis of word order and information structure
- Nuu-chah-nulth language learning
- Initial plans for lexical database for Nuu-chah-nulth morphophonology
    - Interface to FST & tdl
    - Stores data for multiple dialects
- Working on framing dissertation proposal around Thai HPSG

Grammar Matrix extensions

- Adjectives library (TJ Trimble):
    - Predicative, attributive and incorporated (intersective) adjectives
    - Copula-full and copula-less constructions
        - Support for agreement morphology on adjectives and copulas
- Valence-changing lexical rules (Chris Curtis, in progress; see HPSG poster):
    - Building-block approach to causative, passive, applicative, deobjective etc.
    - Separates operations on ARG-ST, semantics, agreement, etc.

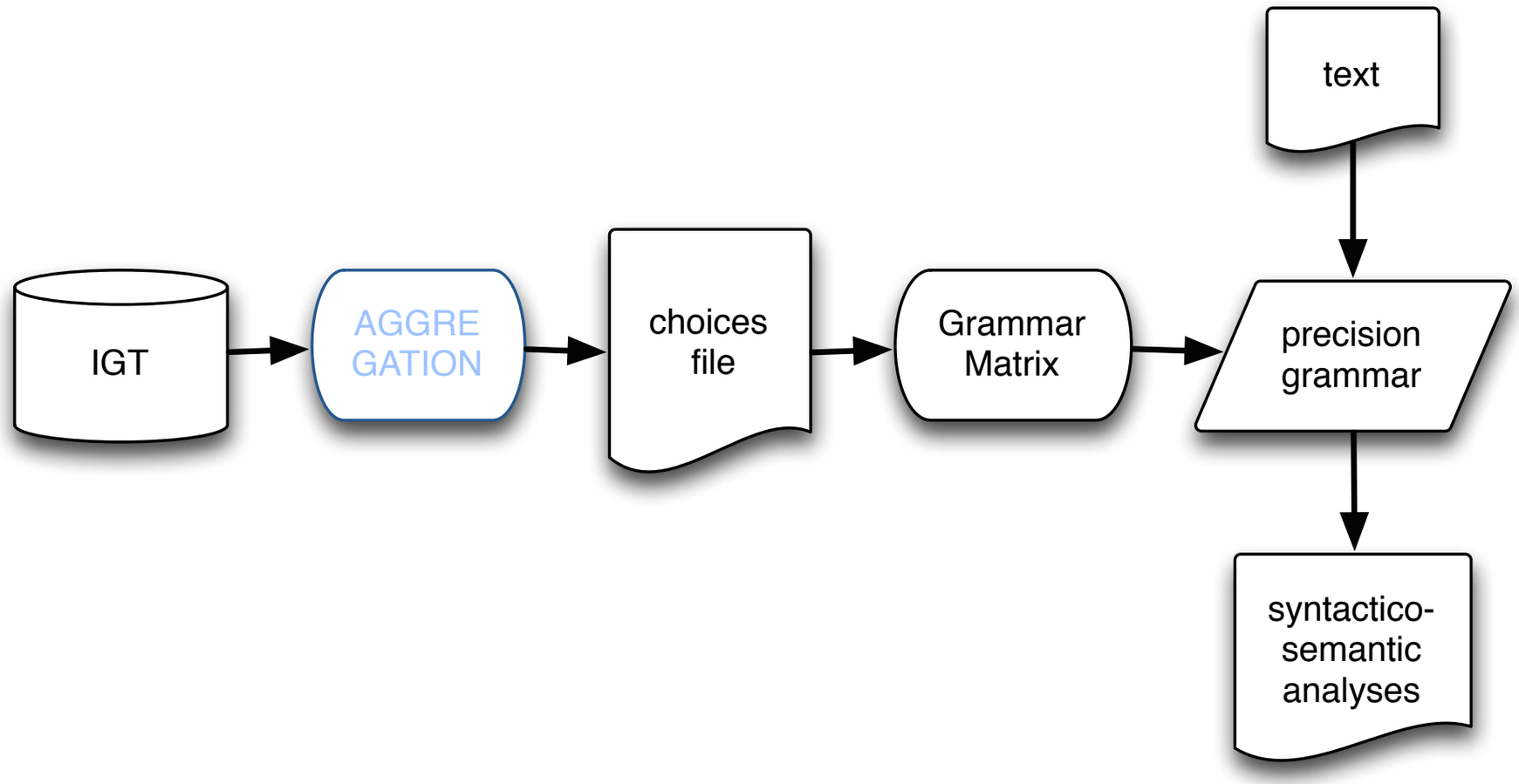
Robustness, Efficiency, Parse Selection Accuracy (Woodley Packard)

- Robustness:
  - reimplemented Jigsaw and got qualitatively similar results but lower scores
  - extrinsic evaluation in the robot control semeval task was discouraging
    - also tried hybrid approach where PCFG proposes edges that the regular parser can incorporate
- Efficiency: added übertagging support to ACE
- Accuracy: self training on wikiwoods shows quite a bit of promise

Semantic Dependency Parsing (Josh Cason)

- built a discriminitive semantic dependency parser based on DeepBank
- comparing results with participants of SemEval 2014 task 8 and 2015 task 18
- worked with interesting algorithms: AdaGrad, Gibbs Sampling, Sample Rank.

AGGREGATION project



Extensions to AGGREGATION

- Xigt compatibility (in progress)

- Ling 575: PNG system, PNG agreement, noun classes, TMA marking, sentential negation, matrix yes-no questions, verb transitivity, coordination strategies, valence change, auxiliaries, case on NPs

- Continuing as MS thesis (Michael Lockwood): General inference methodology to handle

  - inconsistent glossing

  - broaden range of glosses recognized as contentful

  - initial focus on TMA systems

Extensions to AGGREGATION

- Extensions to MOM (Matrix-ODIN Morphology) by Olga Zamaraeva:
    - Xigt-compatible
    - Reduce code to core functionality (PC graph)
    - Add flexibility (can now run on any POS; other small improvements towards more convenient use)
    - Performance improvements
    - PNG feature analysis (started but not completed by Wax)
    - Run on Matsigenka corpus provided by Lev Michael

Congratulations

- to Dr. Anstke Fokkens
- Glenn Slayden, PhC
- Woodley Packard, TJ Trimble (MS theses completed)

